Variceal bleeding before endoscopy — Q73

Which drugs should be given before endoscopy? Give doses and reasons

- Vasoactive drug as soon as variceal bleeding is suspected: terlipressin 2 mg IV 4-hourly (1 mg when bleeding controlled), or octreotide 50 µg bolus then 50 µg/h, for 2–5 days
- Antibiotic prophylaxis: ceftriaxone 1 g/day for up to 7 days – reduces infection, rebleeding and mortality
- IV erythromycin 250 mg 30–120 min before endoscopy – improves visualisation
- Avoid routine FFP/platelets to correct INR (no benefit, raises portal pressure); PPI may be given until endoscopy

Describe your transfusion and resuscitation strategy

- Restrictive transfusion: transfuse at Hb < 7 g/dL, target 7–8 (Villanueva 2013: lower mortality and rebleeding)
- Avoid over-transfusion and excessive fluid (raises portal pressure → rebleeding)
- Airway protection (intubation) if massive haematemesis or encephalopathy before endoscopy

When should endoscopy be performed, and what is done

- Within 12 h of presentation once resuscitated (Baveno VII)
- Endoscopic band ligation for oesophageal varices
- Cyanoacrylate glue injection for gastric (fundal) varices

Variceal bleeding before endoscopy — Q74

What are the rescue options for refractory variceal bleeding

- Balloon tamponade (Sengstaken–Blakemore/Minnesota tube) as a bridge for up to 24 h (after intubation)
- Self-expanding oesophageal metal stent (fewer complications than balloon)
- Rescue TIPS
- Repeat endoscopy if feasible

Define 'pre-emptive TIPS' and its indications

- TIPS placed within 72 h (ideally < 24 h) after initial successful endoscopic control in high-risk patients
- Indications: Child–Pugh C 10–13, or Child–Pugh B > 7 with active bleeding at endoscopy
- Reduces rebleeding and mortality (García-Pagán 2010)

List the complications of balloon tamponade and of TIPS

- Balloon: oesophageal rupture/ulceration, aspiration, airway obstruction (balloon migration)
- Balloon: rebleeding after deflation in ~50%
- TIPS: hepatic encephalopathy, heart failure (raised preload), shunt dysfunction
- TIPS: liver failure, haemolysis, infection

King's College criteria — Q75

State the King's College criteria for transplantation in paracetamol-induced ALF

- Arterial pH < 7.30 (or lactate > 3.0 mmol/L) after adequate fluid resuscitation
- OR all three within 24 h: INR > 6.5 (PT > 100 s), creatinine > 300 µmol/L (3.4 mg/dL), grade 3–4 encephalopathy
- Modified criteria: lactate > 3.5 mmol/L at 4 h, or > 3.0 after resuscitation, add sensitivity
- She meets criteria (pH 7.24; and INR, creatinine and grade 3 encephalopathy)

State the King's College criteria for non-paracetamol ALF

- INR > 6.5 alone
- OR any 3 of: age < 10 or > 40, aetiology non-A non-B/drug reaction, jaundice to encephalopathy > 7 days, INR > 3.5, bilirubin > 300 µmol/L (17.5 mg/dL)
- Limitations: high specificity but lower sensitivity; MELD and lactate also used

Outline her immediate management

- Continue N-acetylcysteine (benefit even late)
- Urgent referral to a transplant centre
- Intubate for grade 3–4 encephalopathy; protect against cerebral oedema
- Treat hypoglycaemia, infection screening, CRRT for AKI/hyperammonaemia

King's College criteria — Q76

Explain why cerebral oedema occurs in ALF and its risk factors

- Ammonia is converted to glutamine in astrocytes → osmotic swelling (cytotoxic oedema)
- Loss of cerebral autoregulation, systemic inflammation increase cerebral blood flow
- Risk factors: ammonia > 150–200 µmol/L, hyperacute presentation, grade 3–4 encephalopathy, young age, AKI, vasopressor need

Outline neuroprotective measures

- Head up 30°, sedation, normocapnia (PaCO2 35–40 mmHg), avoid hypoxia
- Induce hypernatraemia: Na 145–155 mmol/L with hypertonic saline
- Early CRRT to lower ammonia (high-dose, also controls temperature and acid–base)
- Mannitol for surges in ICP; avoid fever; consider ICP monitoring in selected patients

What other organ support does he need

- Haemodynamic: noradrenaline for vasodilatory shock; hydrocortisone if refractory
- Glucose monitoring – hypoglycaemia common; give dextrose
- Infection surveillance; low threshold for antibiotics/antifungals
- Coagulopathy: no routine correction unless bleeding/procedures (INR is a prognostic marker)

Intra-abdominal hypertension and compartment syndrome — Q77

Describe how intra-abdominal pressure is measured

- Intravesical (bladder) pressure via urinary catheter, instilling ≤ 25 mL sterile saline
- Measured in mmHg at end-expiration, supine, transducer zeroed at the mid-axillary line at the iliac crest
- Abdominal muscles relaxed (no contractions)

Define IAH grades and abdominal compartment syndrome, and calculate his abdominal perfusion pressure

- Normal IAP 5–7 mmHg; IAH = sustained IAP ≥ 12 mmHg
- Grades: I 12–15, II 16–20, III 21–25, IV > 25 mmHg
- ACS = sustained IAP > 20 mmHg with new organ dysfunction
- APP = MAP − IAP = 70 − 24 = 46 mmHg (target ≥ 60) → he has ACS (grade III with oliguria and raised airway pressure)

Explain the organ effects of raised IAP

- Renal: renal venous congestion and compression → oliguria/AKI
- Respiratory: raised diaphragm → reduced compliance, higher Pplat, hypoxaemia
- Cardiovascular: reduced venous return and CO; falsely high CVP
- Gut ischaemia (bacterial translocation), raised ICP

Intra-abdominal hypertension and compartment syndrome — Q78

Outline the medical management of intra-abdominal hypertension

- Evacuate intraluminal contents: NG and rectal decompression, prokinetics, enemas
- Evacuate space-occupying lesions: percutaneous drainage of ascites/collections
- Improve abdominal wall compliance: sedation, analgesia, neuromuscular blockade, avoid head-up > 20°
- Optimise fluids: avoid excess, diuretics or ultrafiltration once stable; aim APP ≥ 60

When is surgical decompression indicated, and what are its complications

- Refractory ACS (IAP > 20–25 with organ failure) despite medical measures
- Decompressive laparotomy with temporary abdominal closure (open abdomen, VAC)
- Complications: fluid/protein loss, fistula, infection, ventral hernia, reperfusion syndrome

List the risk factors for intra-abdominal hypertension in ICU patients

- Large-volume fluid resuscitation (> 5 L/24 h), positive fluid balance
- Pancreatitis, abdominal surgery, trauma, ileus, ascites
- Obesity, prone positioning, high PEEP
- Sepsis, burns, massive transfusion

Non-occlusive mesenteric ischaemia — Q79

What is the diagnosis and its pathophysiology

- Non-occlusive mesenteric ischaemia (NOMI)
- Splanchnic vasoconstriction due to low cardiac output, hypovolaemia and vasopressors (and digoxin)
- Mesenteric vasospasm leads to hypoperfusion and bowel infarction without vessel occlusion

Describe the diagnostic findings

- CT angiography: narrowing of SMA branches, irregular ('string of sausages'), pruning, poor bowel wall enhancement
- Lactate is non-specific and rises late; normal lactate does not exclude ischaemia
- Signs of transmural infarction: pneumatosis, portal venous gas, free fluid

Outline the management

- Treat the cause: optimise cardiac output and volume; reduce vasoconstrictors where possible
- Intra-arterial vasodilator (papaverine) via catheter in the SMA during angiography
- Laparotomy if peritonitis or signs of necrosis (resection, second-look surgery)
- Broad-spectrum antibiotics; bowel rest
- Prefer inodilators; avoid digoxin and high vasopressin doses

Non-occlusive mesenteric ischaemia — Q80

Describe the types of acute mesenteric ischaemia

- Arterial embolism (~50%): AF, post-MI; SMA just beyond the middle colic artery
- Arterial thrombosis (~25%): atherosclerosis, ostial; often prior chronic symptoms
- NOMI (~20%): low-flow states, vasopressors
- Mesenteric venous thrombosis (~5–10%): thrombophilia, portal hypertension, malignancy

Which investigation is best and why

- CT angiography (triphasic) urgently – sensitivity/specificity > 90%
- Do not delay CT for a normal lactate or creatinine

Outline the management of embolic AMI

- Immediate unfractionated heparin
- Urgent revascularisation: endovascular (aspiration, stent) or open embolectomy
- Laparotomy to assess viability and resect necrotic bowel; second-look at 24–48 h
- Fluid resuscitation, broad-spectrum antibiotics
- Long-term anticoagulation for AF

Timing of enteral nutrition — Q81

When should enteral nutrition start in ventilated patients? Explain why

- Within 24–48 h of ICU admission once haemodynamically stable (ESPEN 2023, ASPEN 2022)
- Maintains gut barrier integrity and immune function; reduces infection
- Early EN preferred over early PN and over delayed EN

When should early enteral nutrition be delayed

- Uncontrolled shock (escalating vasopressors, rising lactate)
- Uncontrolled life-threatening hypoxaemia, hypercapnia or acidosis
- Active upper GI bleeding; overt bowel ischaemia; high-output fistula; abdominal compartment syndrome
- Gastric aspirate > 500 mL/6 h

Describe the energy and protein targets in the first week

- Start hypocaloric/trophic: ≤ 70% of energy needs in the early phase (days 1–3), increasing to 80–100% after day 3
- Use indirect calorimetry if available; otherwise 20–25 kcal/kg/day
- Protein 1.3 g/kg/day reached progressively (EFFORT Protein: high protein no benefit, possible harm in AKI)
- Avoid overfeeding (hyperglycaemia, increased CO_2 production, liver dysfunction)

Timing of enteral nutrition — Q82

When would you start parenteral nutrition, and what is the evidence

- If EN is contraindicated or not tolerated: start PN within 3–7 days in well-nourished patients (ESPEN)
- Start PN early (asap) if severely malnourished or high nutritional risk
- EPaNIC (2011): early PN (day 2) vs late (day 8) – late PN had fewer infections and faster recovery
- CALORIES (2014): route (EN vs PN) did not affect mortality when calories similar

How do you assess nutritional risk in the ICU

- NUTRIC score (age, APACHE II, SOFA, comorbidities, days in hospital before ICU, IL-6) – ≥ 5 high risk
- NRS-2002 ≥ 5 high risk; GLIM criteria for malnutrition
- Clinical: weight loss, low intake, muscle mass loss (ultrasound/CT)

What is the role of gastric residual volume monitoring

- Routine GRV monitoring is not needed in most patients
- If used, delay EN only when GRV > 500 mL/6 h
- Manage intolerance with prokinetics (erythromycin, metoclopramide) and post-pyloric feeding